Traffic Sign Detection and Recognition (TSDR)

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Link to GitHub Repository

Introduction

Recently, I have noticed a decline in general driving ability on the roads of the GTA, as well as on the internet. Modern safety systems may be able to counteract this trend, making it easier and safer to drive.

The goal of this project is to develop a robust Traffic Sign Detection and Recognition (TSDR) system capable of identifying and classifying road signs. Recognition of traffic signs is a critical component for some safety systems and autonomous vehicles, where misinterpretation can lead to system failure (such as failing to see a "Stop" sign or misreading a "60" speed limit as "80").

Traditional computer vision methods that rely on hand-crafted features such as colour thresholding or edge detection can be susceptible to changes in conditions. Deep learning, specifically Convolutional Neural Networks, will work effectively for this application, since it learns robust feature representations from the data, allowing the model to generalize well to noisy, blurry, or dark environments.

Illustration

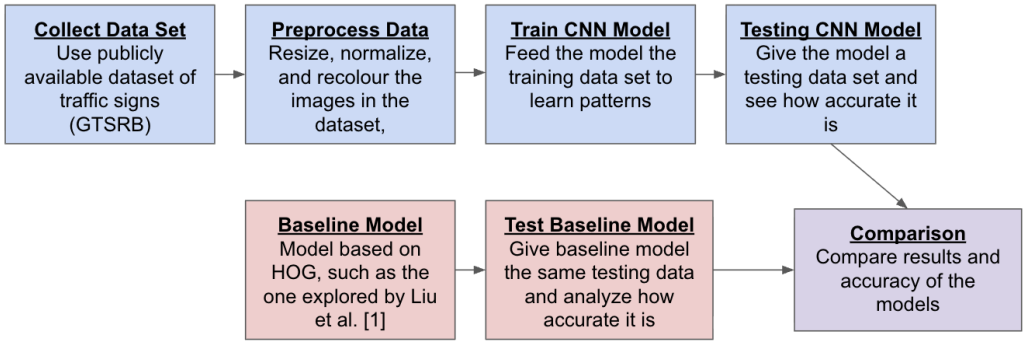

Figure 1: Proposed Traffic Sign Detection and Recognition (TSDR) Model Workflow

Background & Related Work

The following works represent key milestones in the field of traffic sign recognition:

- **Stallkamp et al. [6]:** Introduced the German Traffic Sign Recognition Benchmark (GTSRB), a comprehensive dataset containing over 50000+ samples across 43 classes. This benchmark established the standard for the field by incorporating real-world challenges such as motion blur, harsh lighting variations, and occlusions, providing a rigorous testbed for comparing human and machine performance.
- **Sermanet and LeCun [4]:** Applied Multi-Scale Convolutional Networks (ConvNets) to the GTSRB dataset. Their architecture fed features from the first stage directly to the classifier alongside second-stage features, allowing the model to utilize both local details and global

shapes. This approach achieved an accuracy of 99.17%, surpassing human performance (98.81%) and demonstrating the superiority of deep learning over hand-crafted features.

- **Cireşan et al. [2]:** Utilized Multi-Column Deep Neural Networks (MCDNN), training several CNN's on preprocessed data and averaging their predictions, further pushing the possible accuracy of the task, achieving a record accuracy of 99.46%
- **Zhu et al. [8]:** Explores a fully Convolutional Network to detect traffic signs in large, high-resolution images, focusing on the speed required for real-time autonomous driving.
- **Redmon et al. [3]:** Introduced "You Only Look Once" (YOLO), a unified architecture that treats object detection as a single regression problem rather than a classification task applied to sliding windows. While a general object detection framework, its ability to process images in real-time makes it highly applicable to traffic sign detection where low latency is essential for effective operation of safety features or self driving.

Data Processing

I will use the German Traffic Sign Recognition Benchmark (GTSRB) [6], containing 50,000+ images across 43 classes. To ensure the model is robust, the following will be performed:

- **Balancing:** To account for any significant class imbalances (i.e. many speed limit signs but not many school zone signs), the distribution of the signs will be analyzed and perform oversampling on rare classes to prevent the model from becoming biased toward more frequent signs.
- **Preprocessing:** Images will be resized to a fixed input dimension and normalized to have zero mean and unit variance. The images will also be converted from RGB to the YUV colour space, as suggested by Sermanet and LeCun[4], to separate luminance from chrominance
- **Augmentation:** To improve robustness, a data augmentation pipeline will be made, which randomly applied rotations, blur simulating motion, and brightness adjustments to training images. This will help simulate even more real-world driving conditions that may not be represented in the clean training set.

Architecture

The final model is a multi-scale CNN inspired by Sermanet and LeCun [4], accepting 32×32 RGB images as input. It consists of three convolutional stages, each containing two Conv2d layers (with batch normalization and ReLU activations), followed by a MaxPool2d and a spatial dropout layer (rate 0.25):

- **Stage 1:** $3 \to 32$ channels $\Rightarrow$ output $(B, 32, 16, 16)$
- **Stage 2:** $32 \to 64$ channels $\Rightarrow$ output $(B, 64, 8, 8)$
- **Stage 3:** $64 \to 128$ channels $\Rightarrow$ output $(B, 128, 4, 4)$

The key design choice is a *multi-scale skip connection*: the flattened output of Stage 1 (8,192 features) is concatenated with the flattened output of Stage 3 (2,048 features), yielding a 10,240-dimensional vector. This allows the classifier to simultaneously exploit low-level texture features and high-level abstract representations. The classifier head consists of a fully connected layer (10,240 $\to$ 256), batch normalization, ReLU, dropout (rate 0.5), and a final linear layer ($256 \to 43$) producing class logits. The model was trained with cross-entropy loss (label smoothing 0.1), AdamW (lr $= 10^{-3}$, weight decay $= 10^{-4}$), and a cosine annealing scheduler over 30 epochs.

Baseline Model

Our baseline is an SVM trained on Histogram of Oriented Gradients (HOG) features, as explored by Liu et al. [1]. Images are resized to 32×32, converted to YUV colour space, and HOG features are extracted from the luminance channel (9 orientations, 8×8 pixels per cell, 2×2 cells per block), yielding a 324-dimensional feature vector. An RBF-kernel SVC (C=10, gamma=`scale`)

with standardization is then trained on the full training set of 26,640 samples. This classical pipeline achieves a test accuracy of **81.60%** on the 12,630-sample GTSRB test set, serving as a benchmark to quantify the gains of the deep learning model. Comparing our results with published GTSRB leaderboard results [5] also allows contextualisation against other deep learning architectures.

Quantitative Results

Table 1 summarises test-set accuracy for all evaluated models.

Table 1: Test accuracy on the GTSRB test set (12,630 samples across 43 classes).

Model	**Test Accuracy**
SVM + HOG (baseline)	81.60%
CNN (ours)	98.70%
Sermanet & LeCun [4]	99.17%
Cireşan et al. [2]	99.46%

The CNN achieves a **17.1 percentage-point improvement** over the SVM+HOG baseline and approaches state-of-the-art single-model performance. Across 43 classes, the macro-averaged F1-score is 0.980 and the weighted-average F1-score is 0.987. The hardest class was *Double curve* (66.7% recall), where limited inter-class visual variation led to frequent misclassifications; the easiest classes (*Ahead only*, *Go straight or right*, *Go straight or left*, *End of no passing*, and *End of no passing by vehicles over 3.5t*) all achieved 100% accuracy.

Qualitative Results

Visualising correct and incorrect predictions reveals that the CNN performs robustly across varying lighting and blur conditions. Correct predictions include signs captured under glare, at oblique angles, and with partial occlusion. Misclassifications are concentrated within visually similar subgroups: speed limit signs that share the same red-circle design but differ only in their numeral (e.g., 60 km/h confused with 80 km/h), and warning signs with similar triangular silhouettes (e.g., *Double curve* confused with *Slippery road*). These errors are consistent with the per-class accuracy analysis and motivate future work on auxiliary digit-recognition heads or attention mechanisms.

Evaluate Model on New Data

The GTSRB test split comprises 12,630 images that were held out entirely during training and hyperparameter search, making it a fair proxy for unseen real-world data. The CNN achieves **98.70%** on this held-out test set, close to its best validation accuracy of 99.83%, indicating that the model generalises well and has not overfit to the validation distribution used for early stopping. The small gap between validation and test accuracy (≈ 1.1 pp) is attributable to the validation set being drawn from the same training distribution (via torchvision's `random_split`), while the GTSRB test images were collected independently under a broader range of conditions. Overall, model performance meets expectations for a single-model approach trained from scratch on the GTSRB benchmark [6].

Discussion

The CNN significantly outperforms the SVM+HOG baseline, confirming that learned feature representations are superior to hand-crafted ones for traffic sign recognition under varied conditions. The multi-scale skip connection, concatenating Stage-1 and Stage-3 feature maps before the classifier, captures both fine-grained texture (numerals, arrow shapes) and global shape information simultaneously, which is likely a key contributor to performance. The weighted random sampler and augmentation pipeline (random rotation, motion blur, colour jitter) improved robustness to rare classes and real-world noise respectively.

The primary failure mode is within visually similar subgroups: speed limit signs share an identical template and differ only in numeral, making them inherently confusable at 32×32 resolution. *Double curve* remains the hardest class (66.7% recall), plausibly because it appears rarely in training and is visually similar to other warning signs. Despite these edge cases, the overall result of 98.70% is competitive with literature, falling within 0.47 pp of Sermanet & LeCun [4] and 0.76 pp of Cireşan et al. [2], both of which employed substantially more complex ensembles or multi-column architectures. The result suggests that a clean single-model design with thoughtful data handling is sufficient to approach human-level performance (98.81%) on this benchmark [6].

Ethical Considerations

The model's failure in safety-critical scenarios (e.g., missing a "Stop" sign) poses significant risks to public safety. A deployed TSDR system should never serve as the sole decision-maker in a safety-critical decision, human oversight and redundant sensing are essential. Prolonged reliance on driver-assistance features could also erode a driver's own situational awareness, creating risk if the system fails unexpectedly.

The cameras required for real-time TSDR can capture bystanders, licence plates, and private property, raising data-retention and access concerns analogous to the Tesla privacy incident [7].

Finally, GTSRB contains exclusively German signs; deploying this model in regions with different sign designs (e.g., North America) without domain adaptation or re-training on locally representative data could introduce systematic errors. Adversarial robustness is also an open concern: small physical stickers on signs have been shown to fool CNNs in practice, and this vulnerability must be addressed before any safety-critical deployment.

References

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